

SAANVI REDDY BARADI

United States | +1 (656) 347-6008 | baradisaaanvireddy2003@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

AI Engineer building production-oriented generative AI, LLM, and agentic systems, from RAG pipelines and QLoRA/PEFT fine-tuning to FastAPI services deployed on Docker, Kubernetes, and Terraform with full observability. Also builds the data engineering underneath and communicates findings clearly to stakeholders.

TECHNICAL SKILLS

Generative AI, LLMs & Agentic AI: LangChain (Chains, Agents, LCEL), LangGraph (ReAct, Tool Orchestration, Multi-Agent Workflows), LlamaIndex, QLoRA/PEFT Fine-Tuning (Unsloth), Prompt Engineering, Function Calling, RAG Pipelines, OpenAI API, Anthropic API, Vertex AI (Gemini), Groq (Llama 3.3), LangSmith, LLM Evaluation (Ragas), Groundedness & Safety Evaluation, Guardrails, Hallucination Detection & Mitigation, Responsible AI Practices

Programming & Software: Python, Java, SQL, R, MATLAB, Object-Oriented Programming (OOP), FastAPI, RESTful API Design, Pydantic, pytest, Git/GitHub, CI/CD (GitHub Actions), Docker, Error Handling, Rate Limiting

Vector Search & Embeddings: Pinecone, ChromaDB, FAISS, Milvus, PostgreSQL (pgvector), Hugging Face Transformers & Embeddings (all-MiniLM-L6-v2), Vertex AI Embeddings, Semantic Search, Similarity Retrieval

Data Engineering: Pandas, ETL Pipeline Design, Apache Spark, Kafka, Databricks, Snowflake, BigQuery, PostgreSQL, dbt, Airflow, RapidFuzz (Fuzzy Matching), NoSQL, Data Validation & Quality Assurance, Schema Validation, REST API Integration

Cloud & MLOps: GCP (Vertex AI, Cloud Run, Cloud Logging), AWS EKS, Azure, Azure AKS, Kubernetes, Helm, Terraform, Docker & Compose, Streamlit, Linux, Model Serving, Production Monitoring, Containerization

Machine Learning & Statistics: TensorFlow (Keras), PyTorch, Scikit-learn, XGBoost, SHAP, SMOTE, Supervised & Unsupervised Learning, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Model Evaluation (Precision/Recall, Calibration), Hypothesis Testing

NLP: spaCy, NLTK, TF-IDF, Named Entity Recognition (NER)

Analytics & Visualization: Tableau, Power BI (DAX, PL-300 Certified), Excel, Matplotlib, Seaborn, A/B Testing, Cohort Analysis, RFM Segmentation

EXPERIENCE

AI & Data Analyst – Compass Group / University of South Florida

Jan 2026 – Present

- Partners with stakeholders to translate AI problem statements into technical requirements, evaluation criteria, and experimental objectives.
- Supports data preparation and validation to turn raw operational data into analytics-ready datasets, applying Python, SQL, and statistical analysis to investigate patterns and communicate findings.
- Evaluates AI-assisted workflows by defining measurable outcomes and comparing results across approaches, then prototypes GenAI and RAG workflows covering document processing, embeddings, vector retrieval, and context construction.
- Runs structured experiments comparing prompts, retrieval strategies, and model configurations, and applies human-in-the-loop review to catch unsupported outputs before communicating results to stakeholders.

Project Assistant – University of South Florida

Feb 2025 – Sep 2025

- Designed a Python ETL pipeline that standardized 180+ country-name variations into ISO codes using RapidFuzz, cutting errors by about 90%.
- Automated ingestion, preprocessing, feature engineering, and schema validation across the AI pipeline, cutting manual data-processing effort by roughly 70%.
- Integrated the U.S. Census Bureau's ACS 5-Year API using FIPS-based conditional logic for demographic extraction.
- Built reusable Python modules for data validation, anomaly detection, duplicate identification, and exception handling, with Git-based CI/CD workflows for testing and validation.

Data Science Intern – Accenture

Jun 2023 – May 2024

- Built 15+ SQL reporting pipelines on cognitive and socioeconomic datasets, cutting report-generation time by about 40%.
- Created 8+ Tableau and Power BI dashboards using DAX to translate analysis outputs into executive KPIs.
- Optimized SQL queries on million-row tables through indexing and refactoring for a 40%+ cut in query execution time.
- Designed automated data-quality frameworks that reduced pipeline failures by about 35%.

Research Assistant – Malla Reddy Engineering College for Women

Jun 2021 – Jun 2023

- Built an end-to-end NLP pipeline analyzing 18,617 OSHA severe injury reports, applying TF-IDF and spaCy NER for feature extraction.
- Evaluated SVM, Random Forest, Gradient Boosting, and Logistic Regression with SMOTE to address class imbalance across the report categories.
- Reached 96% accuracy and 0.9867 AUC-ROC with the best-performing model on held-out data.
- Integrated Google Gemini to generate safety recommendations and identify relevant OSHA standards from model outputs, connecting NLP, predictive modeling, and generative AI into one applied research workflow.

PROJECTS

Production-Oriented Agentic AI Platform

LangGraph, FastAPI, pgvector, Ollama, Kubernetes, Helm, Terraform

- Built a hybrid-routing agentic AI assistant with LangGraph that answers over both unstructured policy documents (RAG with pgvector and page-level citations) and structured HR data (text-to-SQL with AST-level guardrails and read-only enforcement), backed by a local Ollama LLM, Redis-based memory and caching, and JWT-authenticated FastAPI endpoints.
- Reached 100% hit-rate and page recall at k=4 on retrieval and 70% text-to-SQL execution accuracy in evaluation, and deployed the full stack with Docker, Kubernetes (Helm charts with an EKS overlay), Terraform IaC, and Prometheus, Grafana, and OpenTelemetry observability, running entirely locally for \$0.
- Built a comprehensive evaluation framework (Ragas for RAG, execution-based scoring for SQL) with multi-layer guardrails covering write-intent detection and grounding checks, wired into GitHub Actions CI/CD with retrieval, SQL, and Ragas evaluation gates on every change.

Multi-Cloud Fine-Tuned LLM Serving Pipeline

QLoRA, PEFT, Unsloth, Pinecone, Ragas, FastAPI, Kubernetes, Terraform

- Fine-tuned Mistral-7B with QLoRA (4-bit LoRA adapters via Unsloth) on a 24K-example customer-support dataset, bringing training loss down from 0.79 to 0.63.
- Added Pinecone-backed RAG with automated Ragas evaluation, and served the model behind a FastAPI and Docker deployment with an automated evaluation gate inside a GitHub Actions CI/CD pipeline.
- Wrote Helm charts and Terraform IaC validated for both AWS EKS and Azure AKS, enabling consistent multi-cloud deployment.

RAG Document Q&A API Service

FastAPI, LangChain, pgvector, Vertex AI, Pytest

- Built an end-to-end RAG service for PDF and text ingestion using Vertex AI embeddings and pgvector retrieval, exposing FastAPI endpoints with Pydantic validation, rate limiting, Cloud Logging, and response latency under 100ms.
- Built a ROUGE-based evaluation framework to score generated answers against source documents, catching hallucination and grounding failures before they reached users.
- Covered ingestion, retrieval, and endpoint logic with a Pytest suite to catch regressions before deployment.

EDUCATION

University of South Florida – Master of Science, Artificial Intelligence and Business Analytics, Tampa, FL (GPA: 3.95/4.0)

Aug 2024 – May 2026

Malla Reddy Engineering College for Women – B.Tech, Computer Science Engineering, Hyderabad, India (GPA: 3.7/4.0)

Jun 2020 – Apr 2024

CERTIFICATIONS

Power BI Data Analyst (PL-300) – Microsoft | Data Science Fundamentals & Introduction to Data Science – IBM | Programming Essentials in Python – Cisco/OpenEDG | Python (Basic) – HackerRank